

		Taking the background count rate into consideration, the reading will be $127 + 24 = 151$.
30.	D	Since the radiation is able to pass through 1 mm of aluminium sheet, it cannot be alpha radiation which can get blocked by paper. From the topic of electromagnetism, we learnt that the radius of the circular motion of a charged particle in a magnetic field is directly proportional to the speed of the particle. $r = \frac{mv}{qB}$ From the figure, we can see that the radius on left side of sheet is smaller than that on right side, thus it is likely that the radiation slowed down upon passing through the sheet. Therefore the path must be from Y to X.